

STAR FOX 64

PRACTICE ROM



BETA USER GUIDE

(!) SD CARD SAVING: NOT WORKING ON EVERDRIVE

SD card save states are in development and do NOT function on EverDrive hardware. RAM save states (D-pad shortcuts) work on all hardware but are cleared on power-off. See the SD Card section for full details.

WHAT IS INCLUDED

SAVE STATES

Save + load your exact game position instantly

CHEATS

Infinite health, bombs, lives, and boost meter

VISUALIZERS

Colored hitboxes, spawn zones, live enemy HP bars

LOADOUT

Laser, bombs, lives, wings, wingmen — editable mid-run

FRAME ADVANCE

Pause and step forward one frame at a time

HUD OVERLAYS

Hit counter, speed, lag frames, charge timing

LEVEL SELECT

Warp to any stage or checkpoint directly

INPUT MACROS

Record and replay button sequences

QUICK START Press **Z** +  during gameplay to open the practice menu.

Thank You

Thank you sincerely for testing the beta version of the Star Fox 64 Practice ROM. This ROM attempts to maintain as much vanilla code as possible while unlocking helpful practice tools similar to those found in other N64 practice ROMs.

A huge thank you to the Star Fox 64 decomp team -- this project would not exist without their incredible work reverse-engineering the original game. And to the HIT64 community: thank you for being so welcoming and for all the help and feedback that has shaped this ROM.

This early on, there are bound to be bugs and crashes. Please send over anything you find and I'll get it fixed and released as a new version at sageraces.com as soon as possible. For future betas, keep an eye on staging.sageraces.com/practice-roms and check the Discord.

(!) SD card save states do NOT work on EverDrive hardware yet. All other features -- RAM save states, frame advance, cheats, HUD overlays, macros, level select -- work normally.

WHAT'S IN THIS ROM

SAVE STATES	Save and instantly reload your exact position and game state at any point.
FRAME ADVANCE	Pause gameplay and step forward one frame at a time for precise analysis.
CHEATS	INF HEALTH, INF BOMBS, INF LIVES, INF BOOST, and auto charge-shot.
HUD OVERLAYS	Hit counter, speed, lag frames, charge timing, and missed input tracking.
VISUALIZERS	Colored hitboxes on enemies, scenery, items, and the player ship.
ENEMY HP BARS	Live health bars on enemies and bosses, sortable and filterable.
LEVEL SELECT	Warp to any stage or checkpoint directly from the practice menu.
LOADOUT EDITOR	Change laser, bombs, lives, rings, health bar, and wingman status mid-run.
INPUT MACROS	Record and replay button sequences for routing analysis.
SD CARD SAVES	(In development) Save states to SD card for persistent run history.

REPORTING BUGS

- Note the level you were on, what you were doing, and what happened.
- Video or screenshot clips are very helpful.
- Send reports directly to the project maintainer.
- Crashes on specific levels, unexpected menu behavior, and broken saves are top priority.

OPENING THE MENU

Press **Z + D-pad Right** at any time during active gameplay to open the radial practice menu.

DURING GAMEPLAY (menu CLOSED)

These D-pad shortcuts ONLY work while the radial menu is closed. Once the menu is open, D-pad directional inputs do nothing.

 + 	Open the radial practice menu
	Pause / unpause frame advance
	Advance one frame (only while paused)
	Save state to the active slot
	Load state from the active slot

WHILE MENU IS OPEN

D-pad directional shortcuts are disabled while the menu is open. Stick controls the menu. Only these controls apply:

	Confirm the highlighted option
	Return to the title screen (hold)
 / 	Cycle the active save slot (shown in HUD)
	Save to SD card (!) NOT WORKING ON EVERDRIVE
 + 	Load from SD card (!) NOT WORKING ON EVERDRIVE

SUBMENU NAVIGATION

After selecting a menu option (e.g. DISPLAY or LOADOUT), a list submenu opens:

	Move cursor up
	Move cursor down
	Decrease / previous value
	Increase / next value
	Toggle option on/off or enter a sub-submenu
	Go back / close submenu

THE RADIAL PRACTICE MENU

Press Z + D-pad Right to open the menu. Tilt the control stick toward any option, then press A to confirm. Options are arranged like a compass -- stick up selects RESTART, stick right selects SAVE, etc.



MENU OPTIONS

RESTART	stick up	Restart the current level from the beginning.
DISPLAY	up-right	Toggle HUD overlays: hit counter, speed, lag frames, input display, minimap, and more.
SAVE	stick right	Open save state manager -- browse slots, save to a specific slot.
LOAD	down-right	Instantly load the most recent state in the active slot.
LEVELS	stick down	Warp to any level or checkpoint.
LOADOUT	down-left	Change laser, bombs, lives, rings, health bar, wings, and wingman status.
CHEATS	stick left	Toggle cheats: INF HP, INF BOMBS, INF LIVES, INF BOOST, auto charge-shot.
SD CARD	up-left	(!) NOT WORKING ON EVERDRIVE -- SD card save state I/O (in development).

SAVE STATES

Save states capture your exact game position, ship state, enemy layout, and level progress. Load them instantly to retry difficult sections without replaying from the start.

QUICK SAVE / LOAD (D-pad, menu closed)

	Save state to the active slot
	Load state from the active slot

SLOT MANAGEMENT (menu open)

 / 	Cycle to the previous / next save slot
SAVE option	Open full save menu -- browse slots, save to a specific slot
LOAD option	Load from the active slot directly

The active slot number is shown in the HUD overlay. Cycle slots with L/R while the menu is open, then save/load with D-pad LEFT/RIGHT during play.

FRAME ADVANCE

Pause the game at any moment and step forward one frame at a time. Useful for analyzing inputs, routing, and hitbox interactions.

	Toggle pause / unpause
	Advance exactly one frame (only while paused)
	Save state (works while paused)
	Load state (works while paused)

Frame advance and save states work together -- pause with D-DOWN, step to the exact frame, then D-LEFT to snapshot that moment.

SD CARD SAVES (!)

(!) SD card saving is NOT functional on EverDrive hardware. The SD CARD option is visible in the menu but saving/loading to SD will silently fail. Use RAM save slots instead -- they work on all hardware but are cleared on power-off.

SD card saves will eventually allow:

- Persistent save states that survive power cycles (RAM saves do not)
- Sharing save files between consoles
- Multiple named save files browsable from the menu

CHEATS

Select CHEATS from the radial menu. A secondary radial dial appears -- tilt the stick toward a cheat and press A to toggle it on or off. Active cheats stay on until toggled off or the game restarts.

INF HP

INF HEALTH

Player cannot take damage. Health bar locked at maximum.

INF BOMB

INF BOMBS

Smart Bomb count never depletes. Fire as many bombs as you want without running out.

INF LIFE

INF LIVES

Lives counter is locked. Game over screens are prevented -- you always continue.

INF BST

INF BOOST

Boost meter never depletes. Boost and brake freely without penalty. Laser never overheats.

AUTO SHOT

AUTO CHARGE SHOT

Automatically fires a charge shot when the charge meter is full, without releasing A.

LOADOUT

Select LOADOUT from the radial menu to customize your ship's equipment mid-run:

LASER	Cycle laser type: SINGLE -> TWIN -> HYPER
BOMBS	Set Smart Bomb count (0-9)
LIVES	Set extra life count
RINGS	Set ring (shield) count
HEALTH	Toggle health bar length: SHORT (3 hits) or LONG (full)
R WING	Toggle right wing: NONE / BROKEN / INTACT
L WING	Toggle left wing: NONE / BROKEN / INTACT
FALCO	Toggle Falco: ALIVE or DOWN
SLIPPY	Toggle Slippy: ALIVE or DOWN
PEPPY	Toggle Peppy: ALIVE or DOWN
EXPERT	Toggle expert mode on / off
HITS	Set current hit count (score counter)
PLANETS	Configure which previous planets count as cleared

DISPLAY OVERLAYS

Select DISPLAY from the radial menu to toggle HUD information overlays. These render over the normal game HUD without affecting gameplay.

STATS submenu

HUD	Master toggle for the entire practice HUD overlay
LAG	Frame lag counter -- frames where the game dropped below 60 fps
SPEED	Current ship speed (base + boost modifier)
CHARGE	Charge shot timing -- shows charge level and lock-on state
MISSED	Counts inputs buffered but not registered by the engine
HITS	Real-time hit count synced with the scoring engine

Other display options

SCENES	Cutscene handling: PLAY (normal) or SKIP (auto-advance)
INPUT	Show on-screen input display (button presses visualized)
MINIMAP	Show a minimap overlay when the game is paused

VISUALIZERS

Inside the DISPLAY submenu, VISUALIZERS draws colored overlays on in-game objects to show hitbox boundaries and spawn zones.

HITBOXES submenu

HITBOXES	Master toggle for all hitbox visualization
ACTORS	Hitboxes on enemies and bosses
SCENERY	Hitboxes on level geometry and obstacles
ITEMS	Hitboxes on collectible items (rings, bombs, etc.)
PLAYER	The player ship's own hitbox
FLASH	Flash hitboxes on hit detection events

SPAWN ZONES submenu

SPAWN ZONES	Master toggle for spawn zone visualization
ENEMIES	Enemy spawn trigger zones (red)
ITEMS	Item spawn trigger zones (green)
SCENERY	Scenery spawn trigger zones (blue)

ENEMY HP DISPLAY

The ENEMY HP submenu (inside VISUALIZERS) shows live health bars:

SHOW	Enable enemy HP bar overlay
SORT	Sort bars by: NEAREST enemy or HIGH HP first
MIN HP	Hide enemies below this HP threshold (or OFF)
FILTER	Show ALL enemies or BOSSES only
MODELS	ON = normal 3-D models, OFF = ghost mode (models hidden)

INPUT MACROS

Input macros let you record a sequence of button inputs and replay them with perfect timing. Useful for routing analysis. Access via the MACRO option inside the DISPLAY submenu.

RECORDING

RECORD: ARMED	Armed -- will auto-start on next input, or at boot if SAVE START is ON
RECORD: START	Active -- every button press is being captured
RECORD: STOP	Stop recording and save the macro

PLAYBACK

PLAY: START	Begin playing back the recorded macro
PLAY: STOP	Stop playback
REWIND	Rewind playback to the beginning
TRIM	Trim the macro to the current playback position

OPTIONS

SAVE START	ON = recording auto-arms at game startup (TAS-style)
LOOP	ON = macro loops continuously after reaching the end
LEN	Shows recorded macro length in frames and seconds

Macros are stored in RAM and cleared on power cycle. They work best with save states -- load a state, arm the macro, replay from the exact same position.

LEVEL SELECT

Select LEVELS from the radial menu to warp directly to any stage or checkpoint. Navigate the planet map the same way as the in-game end-of-level route selection.

The level select warps to the chosen stage starting checkpoint. Wingman status and planet clear flags can be adjusted in the LOADOUT menu.

(!) SD CARD SAVING NOT AVAILABLE

SD card save states are in active development and DO NOT work on EverDrive hardware. The SD CARD option appears in the menu, but any save or load will silently fail. Do not rely on SD card saves for practice sessions.

CURRENT STATUS

- SD card interface: implemented and visible in the menu
- EverDrive X7 / X8 support: written but UNVERIFIED on real hardware
- Status: not safe to use -- data loss or crash possible
- RAM save states are fully functional -- use those instead (cleared on power-off)

WHAT SD SAVES WILL DO (WHEN READY)

- Save states to the SD card inside your EverDrive -- no Controller Pak needed
- States persist across power cycles
- Multiple save files per game, browsable from the menu
- Shareable save files -- copy to PC and share with other runners

IN THE MEANTIME: RAM SAVES

RAM save states work fully right now (cleared on power-off):

- D-PAD LEFT to save, D-PAD RIGHT to load -- instant, every frame
- Multiple slots -- cycle with L/R while the menu is open
- Works alongside frame advance -- save/load while paused
- Fast and reliable on all hardware configurations

For updates on SD card support, check staging.sageraces.com. When SD saving goes live, a new ROM version will be released at sageraces.com.

TIPS & NOTES

- The practice menu can be opened on any frame, even the first frame of a level.
- All cheats and overlays are per-session. Resetting the N64 restores vanilla defaults.
- Save states include the current hit count -- snapshot mid-level scores.
- Cutscene skip (SCENES: SKIP) auto-advances through level intros and dialog.
- The auto charge shot cheat fires on charge completion without releasing A.
- Lag frame counter shows how often the engine dropped a frame -- useful for routing.
- Hitbox visualization works in all render modes with frame advance for precise analysis.